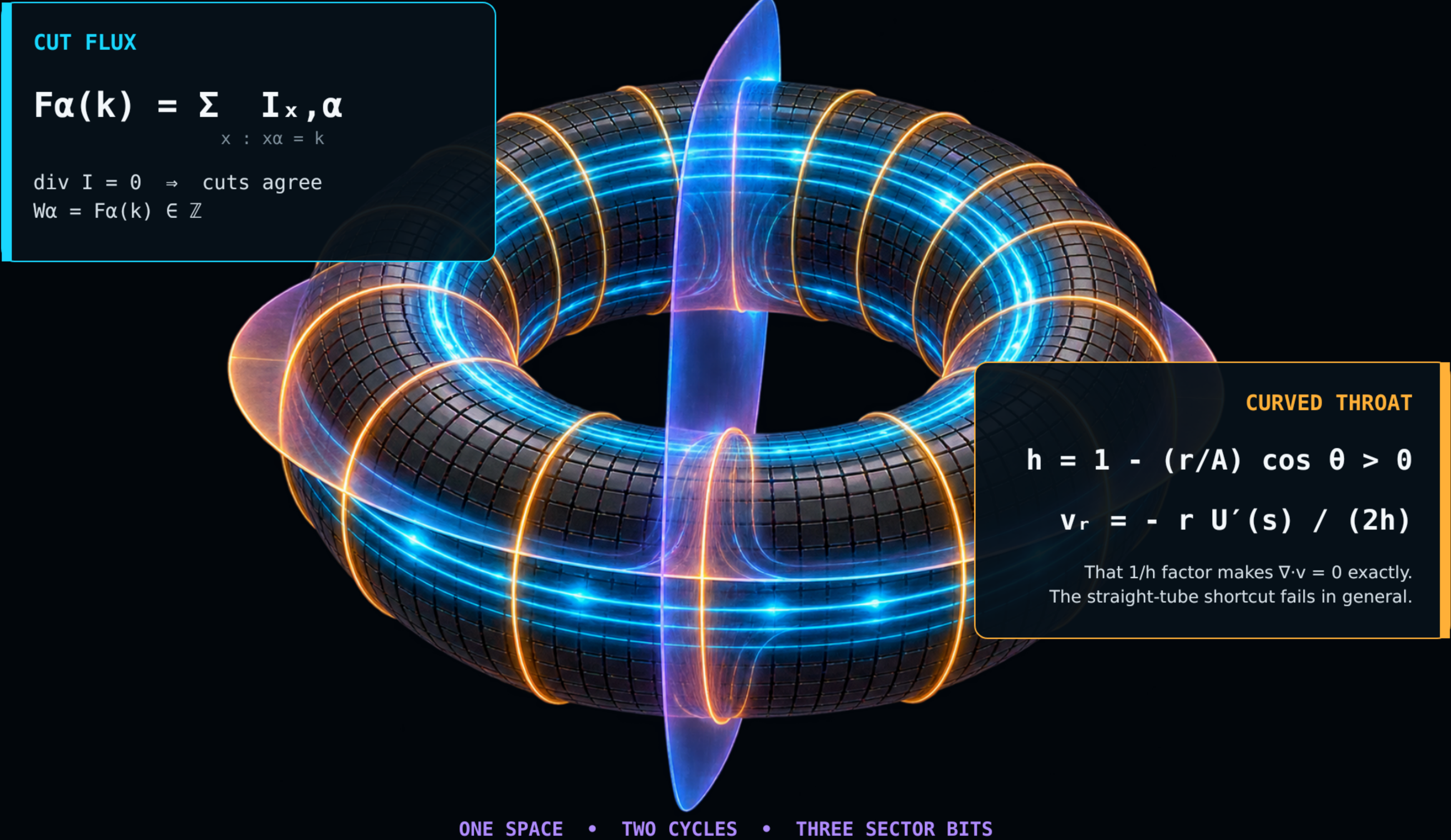


TOROIDAL GEOMETRY WHAT CHANGED TODAY

Cut flux is now the winding invariant. Curvature is built into the throat flow.
A fully specified Monte Carlo reference kernel now exists.



RECTANGULAR TORUS

DIAGONAL ROUTING

Subgroup index = $\text{gcd}(2, M, N)$.
Two sheets iff M and N are both even.

If either period is odd, $(x+y) \bmod 2$ is not globally defined.

REFERENCE DYNAMICS

A REAL KERNEL

$X = (I, M, q)$ on a $2 \times 2 \times 2$ torus.
71 weighted microticks per sweep.

Symmetric proposals + Metropolis give detailed balance; full-state reachability and identity close the kernel proof.

PHASE BRIDGE

13 → 39 coverings are explicit. 39-point quadrature is exact for input modes $|m| \leq 12$ - not globally.

PROVED

cut = winding
curved $\nabla \cdot v = 0$
detailed balance
reachability + aperiodicity

EXECUTABLE

chain-complex validator
64 routing rectangles
modular current checks
kernel contract audit

NOT YET RUN

production sampler
warmup / mixing
Q2 / equilibrium
empirical throat test

NO MYSTICISM • EXACT MAPS • EXECUTION STAYS SEPARATE